Team Leader: Ho Nguyen Phu

Question 1

We have $Z \sim N(0,1)$, so this is Standard Normal Distribution.

(a)

$$\begin{aligned} P(Z \leq 1.34) &= \Phi(1.34) \\ &= \int_{-\infty}^{1.34} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.9099 \end{aligned}$$

(b)

$$\begin{aligned} P(Z \geq -0.22) &= 1 - P(Z \leq -0.22) \\ &= 1 - \Phi(-0.22) \\ &= 1 - (1 - \Phi(0.22)) \\ &= \Phi(0.22) \\ &= \int_{-\infty}^{0.22} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= 0.5871 \end{aligned}$$

(c)

$$\begin{aligned} P(-2.19 \leq Z \leq 0.43) &= \Phi(0.43) - \Phi(-2.19) \\ &= \int_{-\infty}^{0.43} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{-2.19} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.6521 \end{aligned}$$

(d)

$$\begin{aligned} P(0.09 \leq Z \leq 1.76) &= \Phi(1.76) - \Phi(0.09) \\ &= \int_{-\infty}^{0.43} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{0.09} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.4249 \end{aligned}$$

(e)

$$\begin{aligned}
P(|Z| \leq 0.38) &= P(-0.38 \leq Z \leq 0.38) \\
&= \Phi(0.38) - \Phi(-0.38) \\
&= \int_{-\infty}^{0.38} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{-0.38} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&\approx 0.296
\end{aligned}$$

(f)

$$\begin{aligned}
P(|Z| \geq 0.91) &= 1 - P(|Z| \leq 0.91) \\
&= 1 - P(-0.91 \leq Z \leq 0.91) \\
&= 1 - (\Phi(0.91) - \Phi(-0.91)) \\
&= 1 - (\int_{-\infty}^{0.91} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{-0.91} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx) \\
&\approx 0.3628
\end{aligned}$$

(g)

The value of x for which $P(Z \leq x) = 0.55$

$$\begin{aligned}
&P(Z \leq x) = 0.55 \\
&\Leftrightarrow \Phi(x) = 0.55
\end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.125661$

(h)

The value of x for which $P(Z \geq x) = 0.72$

$$\begin{aligned}
&P(Z \geq x) = 0.72 \\
&\Leftrightarrow 1 - P(Z \leq x) = 0.72 \\
&\Leftrightarrow P(Z \leq x) = 0.28 \\
&\Leftrightarrow \Phi(x) = 0.28
\end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx -0.582841$

(i)

The value of x for which $P(|Z| \leq x) = 0.31$

$$\begin{aligned}
&P(|Z| \leq x) = 0.31 \\
&\Leftrightarrow P(-x \leq Z \leq x) = 0.31 \\
&\Leftrightarrow \Phi(x) - \Phi(-x) = 0.31 \\
&\Leftrightarrow \Phi(x) - (1 - \Phi(x)) = 0.31 \\
&\Leftrightarrow 2\Phi(x) - 1 = 0.31 \\
&\Leftrightarrow \Phi(x) = 0.655
\end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.398855$

(j)

The value of x for which $P(|Z| \geq x) = 0.42$

$$\begin{aligned}
&P(|Z| \geq x) = 0.42 \\
&\Leftrightarrow 1 - P(|Z| \leq x) = 0.42 \\
&\Leftrightarrow P(|Z| \leq x) = 0.58 \\
&\Leftrightarrow P(-x \leq Z \leq x) = 0.58 \\
&\Leftrightarrow \Phi(x) - \Phi(-x) = 0.58 \\
&\Leftrightarrow \Phi(x) - (1 - \Phi(x)) = 0.58 \\
&\Leftrightarrow 2\Phi(x) - 1 = 0.58 \\
&\Leftrightarrow \Phi(x) = 0.79
\end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.80642$

Question 2

Suppose we have $X \sim N(10, 2)$
That mean $\mu = 10$ and $\sigma = \sqrt{2}$

(a)

$$\begin{aligned} P(X \leq 10.34) &= \Phi(\frac{10.34 - 10}{\sqrt{2}}) \\ &= \int_{-\infty}^{\frac{10.34-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.5950 \end{aligned}$$

(b)

$$\begin{aligned} P(X \geq 11.98) &= 1 - P(X \leq 11.98) \\ &= 1 - \Phi(\frac{11.98 - 10}{\sqrt{2}}) \\ &= \int_{-\infty}^{\frac{11.98-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.0807 \end{aligned}$$

(c)

$$\begin{aligned} P(7.67 \leq X \leq 9.90) &= \Phi(\frac{9.90 - 10}{\sqrt{2}}) - \Phi(\frac{7.67 - 10}{\sqrt{2}}) \\ &= \int_{-\infty}^{\frac{9.90-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{\frac{7.67-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.4221 \end{aligned}$$

(d)

$$\begin{aligned} P(10.88 \leq X \leq 13.22) &= \Phi(\frac{13.22 - 10}{\sqrt{2}}) - \Phi(\frac{10.88 - 10}{\sqrt{2}}) \\ &= \int_{-\infty}^{\frac{13.22-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{\frac{10.88-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.2555 \end{aligned}$$

(e)

$$\begin{aligned}
P(|X-10| \leq 3) &= P(-3 \leq X-10 \leq 3) \\
&= P(7 \leq X \leq 13) \\
&= \Phi(\frac{13-10}{\sqrt{2}}) - \Phi(\frac{7-10}{\sqrt{2}}) \\
&= \int_{-\infty}^{\frac{13-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{\frac{7-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&\approx 0.9661
\end{aligned}$$

(f)

$$\begin{aligned}
P(|X-10| \geq 8) &= 1 - P(|X-10| \leq 8) \\
&= 1 - P(-8 \leq X-10 \leq 8) \\
&= 1 - P(2 \leq X \leq 18) \\
&= 1 - (\Phi(\frac{18-10}{\sqrt{2}}) - \Phi(\frac{2-10}{\sqrt{2}})) \\
&= 1 - (\int_{-\infty}^{\frac{18-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx - \int_{-\infty}^{\frac{2-10}{\sqrt{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx) \\
&\approx 0.000000015
\end{aligned}$$

(g)

Find x such that $P(X \leq x) = 0.81$

$$\begin{aligned}
P(X \leq x) &= 0.81 \\
\Leftrightarrow \Phi(\frac{x-10}{\sqrt{2}}) &= 0.81
\end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.877896$. So:

$$\frac{x-10}{\sqrt{2}} = 0.877896 \Leftrightarrow x \approx 11.24153$$

(h)

Find x such that $P(X \geq x) = 0.04$

$$\begin{aligned} P(X \geq x) &= 0.04 \\ \Leftrightarrow 1 - P(X \leq x) &= 0.04 \\ \Leftrightarrow P(X \leq x) &= 0.96 \\ \Leftrightarrow \Phi(\frac{x-10}{\sqrt{2}}) &= 0.96 \end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 1.750686$. So:

$$\frac{x-10}{\sqrt{2}} = 1.750686 \Leftrightarrow x \approx 12.47584$$

(i)

Find x such that $P(|X - 10| \geq x) = 0.63$

$$\begin{aligned} &P(|X-10| \geq x) && = 0.63 \\ &\Leftrightarrow 1 - P(|X-10| \leq x) && = 0.63 \\ &\Leftrightarrow P(|X-10| \leq x) && = 0.37 \\ &\Leftrightarrow P(-x \leq X - 10 \leq x) && = 0.37 \\ &\Leftrightarrow P(-x+10 \leq X \leq x+10) && = 0.37 \\ &\Leftrightarrow \Phi(\frac{x+10-10}{\sqrt{2}}) - \Phi(\frac{-x+10-10}{\sqrt{2}}) && = 0.37 \\ &\Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) - \Phi(\frac{-x}{\sqrt{2}}) && = 0.37 \\ &\Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) - (1 - \Phi(\frac{x}{\sqrt{2}})) && = 0.37 \\ &\Leftrightarrow 2\Phi(\frac{x}{\sqrt{2}}) - 1 && = 0.37 \\ &\Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) && = 0.685 \end{aligned}$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.48173$. So:

$$\frac{x}{\sqrt{2}} = 0.48173 \Leftrightarrow x \approx 0.68126$$

(j)

Find x such that $P(|X-10| \leq x) = 0.44$

$$
\begin{aligned}
& P(|X-10| \leq x) && = 0.44 \\
& \Leftrightarrow P(-x \leq X-10 \leq x) && = 0.44 \\
& \Leftrightarrow P(-x+10 \leq X \leq x+10) && = 0.44 \\
& \Leftrightarrow \Phi(\frac{x+10-10}{\sqrt{2}}) - \Phi(\frac{-x+10-10}{\sqrt{2}}) && = 0.44 \\
& \Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) - \Phi(\frac{-x}{\sqrt{2}}) && = 0.44 \\
& \Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) - (1 - \Phi(\frac{x}{\sqrt{2}})) && = 0.44 \\
& \Leftrightarrow 2\Phi(\frac{x}{\sqrt{2}}) - 1 && = 0.44 \\
& \Leftrightarrow \Phi(\frac{x}{\sqrt{2}}) && = 0.72
\end{aligned}
$$

By using Casio 580VN-X, Menu - 7 - 3 to use Inverse Normal function. We calculate and find out inverse $x \approx 0.58284$. So:

$$\frac{x}{\sqrt{2}} = 0.58284 \Leftrightarrow x \approx 0.82426$$

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Date

3) $X \sim N(\mu, \sigma^2)$

We have:

- $P(X \leq 5) = 0.8$

$\Rightarrow \Phi\left(\frac{5-\mu}{\sigma}\right) = 0.8$

$\Rightarrow \frac{5-\mu}{\sigma} \approx 0.8416$

$\Rightarrow 0.8416\sigma + \mu \approx 5 \quad (1)$

- $P(X \geq 0) = 0.6$

$\Rightarrow P(X < 0) = 0.4$

$\Rightarrow \Phi\left(\frac{0-\mu}{\sigma}\right) = 0.4$

$\Rightarrow \frac{0-\mu}{\sigma} \approx -0.2533$

$\Rightarrow -0.2533\sigma + \mu \approx 0 \quad (2)$

From (1) and (2) we have:

$$\begin{cases} 0.8416\sigma + \mu \approx 5 \\ -0.2533\sigma + \mu \approx 0 \end{cases}$$

$$\Rightarrow \begin{cases} \sigma \approx 4.5666 \\ \mu \approx 1.1567 \end{cases}$$

$\Rightarrow \sigma^2 \approx 20.8541$

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4. Let X be the random variable of the thickness of glass sheets.

$\Rightarrow X \sim N(3, 0.12^2)$

a) $P(X > 3.2)$

$= 1 - P(X \leq 3.2)$

$= 1 - \int_{-\infty}^{3.2} \frac{1}{0.12\sqrt{2\pi}} e^{\frac{-(x-3)^2}{2 \times 0.12^2}} dx$

≈ 0.0478

b) $P(X < 2.7)$

$= \int_{-\infty}^{2.7} \frac{1}{0.12\sqrt{2\pi}} \times e^{\frac{-(x-3)^2}{2 \times 0.12^2}} dx$

≈ 0.0062

c) $P(3-c \leq X \leq 3+c) = 0.99$

$\Rightarrow P(-c \leq X-3 \leq c) = 0.99$

$\Rightarrow P(|X-3| \leq c) = 0.99$

$\Rightarrow 2P(X-3 \leq c) - 1 = 0.99$

$\Rightarrow P(X \leq c+3) = 0.995$

$\Rightarrow \Phi\left(\frac{c+3-3}{0.12}\right) = 0.995$

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By using Casio 580VN-X, Menu-7-3 to use Inverse Normal function. We calculate and find out inverse: $\frac{c}{0.12} \approx 2,5758 \Rightarrow c = 0.3091$

d) Suppose that the three glasses are independent, we have:

$$\mu_T = 3 \times \mu = 9 \text{ (mm)}$$
$$\sigma_T^2 = 3 \times \sigma^2 = 0.0432 \text{ (mm}^2\text{)}$$
$$\sigma_T = \sqrt{3} \times \sigma = \frac{3\sqrt{3}}{25} \text{ (mm)}$$

We have a new normal distribution with $T \sim N(9, 0.0432)$

Moreover, the total thickness is more than 9.5 mm

$\Rightarrow P(T > 9.5)$

$$= 1 - P(T \leq 9.5)$$

$$= 1 - \int_{-\infty}^{9.5} \frac{1}{\frac{3\sqrt{3}}{25} \times \sqrt{2\pi}} e^{\frac{-(x-9)^2}{2 \times 0.0432}} dx$$

$$\approx 0.0081$$

e) Suppose that the seven glasses are independent, we have:

$$\mu_S = 7 \times \mu = 21 \text{ (mm)}$$
$$\sigma_S^2 = 7 \times \sigma^2 = 0.1008 \text{ (mm}^2\text{)}$$
$$\sigma_S = \sqrt{7} \times \sigma = \frac{3\sqrt{7}}{25} \text{ (mm)}$$

We have a new normal distribution with $S \sim N(9, 0.1008)$

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The probability that the average height of the seven glasses is more than 3.1 mm:

$$P\left(\frac{S}{7} < 3.1\right)$$

$$= P(S < 21.7)$$

$$= \int_{-\infty}^{21.7} \frac{1}{\frac{3\sqrt{7}}{25} \times \sqrt{2\pi}} e^{-\frac{(x-21)^2}{2 \times 0.1008}} dx$$

$$\approx 0.9863$$

5) Let X be the random variable of the weights of bags.

$$\Rightarrow X \sim N(\mu, 0.05^2)$$

We have:

$$P(X < 10) = 0.01$$

$$\Rightarrow \Phi\left(\frac{10-\mu}{0.05}\right) = 0.01$$

$$\Rightarrow \frac{10-\mu}{0.05} \approx -2.3263$$

By using Casio 580VN-X, Menu-7-3 to use Inverse Normal function. We calculate and find out inverse $\mu \approx 10.1163$

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Question 6

Let X be the random variable of the return of the company. We have an investment in company A has an expected return of \$30,000 with a standard deviation of \$4000.

$$X \sim N(30000, 4000^2)$$

The Probability of returning at least \$25,000 is:

$$\begin{aligned} P(X \geq 25000) &= 1 - P(X \leq 25000) \\ &= 1 - \Phi(\frac{25000 - 30000}{4000}) \\ &= 1 - \int_{-\infty}^{\frac{25000-30000}{4000}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &\approx 0.8944 \end{aligned}$$

So probability is 89.44%